

Vinod Gurram

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PROFESSIONAL SUMMARY

Security is strongest when tested from an attacker's perspective. I break systems to identify weaknesses, then strengthen them through penetration testing, threat detection, and incident response across cloud, network, and endpoint environments. Translate offensive findings into SIEM-driven detection, automated remediation, and defensive controls. Use scripting and machine learning-based anomaly detection to reduce response time and improve security posture.

WORK EXPERIENCE

DREAMLINEAI

Sebring, Florida, USA

Security Analyst

Mar 2026 - Present

- Monitored and analyzed security alerts from Splunk SIEM, AWS CloudTrail, and endpoint logs, triaging 150–250 daily alerts and reducing false-positive escalation rate by 28% through improved correlation rules.
- Investigated security events across cloud and application layers, including API abuse attempts, suspicious IAM activity, and anomalous login patterns, improving detection accuracy by 32%.
- Built Python-based automation scripts to correlate CloudTrail, VPC Flow Logs, and application logs, reducing Mean Time to Triage (MTTT) by 30%.
- Assisted in incident response workflows by supporting containment actions such as disabling compromised IAM keys, isolating EC2 instances, and blocking malicious IPs via WAF rules, reducing average containment time by 25%.
- Conducted initial vulnerability validation from scanner outputs (Snyk / Nessus) and helped track remediation across engineering teams, improving closure rate of critical findings by 35% within SLA timelines.
- Participated in weekly threat review meetings, mapping observed attack patterns to MITRE ATT&CK techniques for improved detection coverage.

MONTCLAIR STATE UNIVERSITY

Montclair, NJ, USA

Graduate Assistant

Jan 2025 – Dec 2025

- Designed and delivered AI-driven cybersecurity labs for 50+ students, integrating ML models with network datasets and improving lab completion rates by 30%.
- Guided students in building and evaluating ML models using PyTorch, TensorFlow, and scikit-learn, achieving 90%+ accuracy in anomaly detection experiments on curated network datasets.
- Taught students IP networking concepts, guiding them in configuring network equipment (such as switches and routers) and troubleshooting system-level issues in a lab environment.

CKOINECT

Hyderabad, India

Cloud Security Engineer

Nov 2022 - Nov 2023

- Engineered and maintained cloud security monitoring across AWS environments using Splunk, AWS CloudTrail, and CrowdStrike Falcon, ingesting and correlating 150–250 daily alerts to detect threats including unauthorized API calls, privilege escalations, and lateral movement attempts.
- Designed and enforced least-privilege IAM policies and RBAC controls across AWS accounts and EKS clusters, reducing over-permissioned roles by 40% and preventing unauthorized cross-account resource access.
- Detected and responded to data exfiltration attempts by monitoring S3 bucket access patterns, VPC Flow Logs, and GuardDuty findings; identified anomalous large-volume data transfers and blocked exfil paths via bucket policies and SCPs, improving detection accuracy by 32%.
- Built Python automation pipelines to ingest and correlate CloudTrail events, CrowdStrike EDR telemetry, VPC Flow Logs, and EKS audit logs into Splunk dashboards, reducing mean time to triage by 30% and enabling real-time container threat visibility.
- Executed containment actions for compromised AWS workloads including revoking IAM credentials, quarantining EC2 and EKS pods via network policy enforcement, blocking malicious IPs via WAF, and rotating exposed secrets in AWS Secrets Manager, reducing incident containment time by 25%.
- Hardened EKS cluster security posture by implementing Pod Security Admission policies, enforcing RBAC with namespace-scoped roles, enabling audit logging, and integrating Falco for runtime threat detection across containerized workloads.

CKOINECT

Hyderabad, India

Cloud Security Intern

April 2022 - Nov 2022

- Deployed and secured containerized workloads on Amazon EKS, configuring namespace-level RBAC, network policies, and Pod Security Standards to enforce least-privilege container execution and restrict inter-pod communication.
- Supported incident response and root-cause analysis by correlating CrowdStrike Falcon EDR telemetry with AWS CloudTrail and GuardDuty findings; performed automated IOC enrichment and log parsing with Python, reducing manual investigation effort by 20+ hours per month.
- Implemented IAM policy guardrails using AWS SCPs and Permission Boundaries to prevent privilege escalation and unauthorized resource access; monitored for data exfiltration indicators including unusual S3 GetObject patterns, cross-region data transfers, and anomalous CloudTrail API call volumes.
- Developed 15+ SOPs and runbooks for EKS security incidents, IAM misconfigurations, and container escape scenarios, standardizing response procedures and reducing mean time to resolution for recurring cloud security events.
- Monitored AWS network security controls including Network Firewall, WAF, and Security Groups; forwarded VPC Flow Logs and DNS query logs to Splunk for real-time anomaly detection, identifying and blocking C2 beacon traffic and port scan activity across

- cloud VPCs.
- Collaborated with DevOps and platform teams to embed security into AWS architecture design, including VPC segmentation, private EKS endpoint configurations, ECR image scanning, and Terraform-based enforcement of security group and IAM baseline controls.

EDUCATION

MONTCLAIR STATE UNIVERSITY

Montclair, NJ

Masters in Cyber Security - 3.8/4.0

Jan 2024 – Dec 2025

MALLA REDDY COLLEGE OF ENGINEERING AND TECHNOLOGY

INDIA

Bachelors in Computer Science - 3.1/4.0

Jun 2019 – May 2023

PROJECTS

AWS CLOUD SECURITY MONITORING DASHBOARD WITH REAL-TIME THREAT DETECTION

- Developed a real-time security monitoring dashboard using AWS services (CloudTrail, GuardDuty, VPC Flow Logs) integrated with SIEM tools like Splunk or open-source ELK Stack, enabling proactive detection of threats such as unauthorized API calls and anomalous network traffic in a simulated multi-account environment.
- Deployed and monitored a Kubernetes cluster (EKS/Minikube) using a Prometheus/Grafana stack to visualize real-time health and performance metrics, utilizing failure simulations to optimize system resilience.
- Implemented automated incident response workflows with Python scripting and AWS Lambda functions to triage alerts, analyze IAM privilege escalations (e.g., detecting iam:PassRole abuse via GuardDuty findings), and trigger remediation actions like isolating compromised EC2 instances—reducing simulated response time by 40%.
- Configured IPS/IDS rules via AWS Network Firewall and WAF to monitor and block web exploits in VPCs, while generating compliance reports on security events to ensure alignment with NIST frameworks and regulatory standards.
- Incorporated DevOps practices by securing containerized applications (Docker/ECR) through automated vulnerability scanning in CI/CD pipelines, demonstrating hands-on tool management for firewalls, EDR equivalents, and endpoint protection in cloud setups.

MITIGATION OF IOT DDOS ATTACKS USING MACHINE LEARNING

- Engineered a high-accuracy (98%) machine learning-driven system to detect and mitigate DDoS attacks on IoT devices strengthening security against vulnerabilities in weak authentication and protocols.
- Trained advanced ML models (Extreme Gradient Boosting, Random Forest, MLP) using 60+ different types of DDoS attack packets, integrating real-time detection via Flask Socket IO to cut response time by 50%.
- Developed and deployed a full-stack security solution with a React.js dashboard and Flask backend, enabling live attack monitoring on a Raspberry Pi for resource-efficient IoT defence.
- Evaluated models using precision-recall curves and confusion matrices on imbalanced datasets, achieving F1-scores above 97% for minority attack classes and demonstrating robustness against evolving DDoS variants.

SECURE FILE SHARING APPLICATION

- Designed and developed a centralized secure file-sharing application using AES (Advanced Encryption Standard) for robust data encryption.
- Implemented key-based authentication, preventing unauthorized access and reducing data breaches by 90%.
- Ensured end-to-end encryption, maintaining 100% data integrity and confidentiality.
- Integrated secure key derivation (e.g., PBKDF2 with salt) and secure random key generation to protect against brute-force and dictionary attacks, while supporting multi-user access control via role-based permissions.
- Added secure file upload/download over HTTPS with integrity verification (HMAC or checksums), enabling tamper-proof transmission and audit logging of access events for compliance and incident response.

SKILLS

Operating System: Linux/Unix, Windows.

Database/Cloud/Frameworks: SQL, MongoDB, AWS Security, Terraform, Azure Security, Docker, Jenkins, Kubernetes.

Security Tools and Technologies: Nessus, Nmap, Penetration Testing, CrowdStrike Falcon, Falco, AWS GuardDuty, AWS Security Hub,

VLANs, Subnetting, Network Security, TCP/IP, Tenable, Metasploit, IPS/IDS, Splunk, EDR, XDR, Cryptography, Incident Response, Vulnerability Management, Cloud Security, EKS Security, Container Security, RBAC, IAM Policies, Data Exfiltration Detection, AWS SCPs, Pod Security Admission, Active Directory, NIST Framework, KQL, SOC2, ISO 27001.

Languages: Python, JavaScript, PowerShell, C, C++, Bash, Java, Go

CERTIFICATIONS

- [CompTIA Security+](#)
- **Certified AI Security Professional (In Progress)**

ADDITIONAL INFORMATION:

- Write technical [Medium](#) blogs whenever time permits, sharing insights and solutions related to cybersecurity.
- Actively participate in **Hack the Box** challenges, solved 20+ vulnerable machine challenges to enhance practical penetration testing skills.
- Actively engage in CTF challenges to refine exploit development and vulnerability analysis.